

Agile Methodology

What is Agile?

Agile is a project management and software development approach that focuses on delivering small, working pieces of software regularly. It values collaboration, adaptability, and continuous feedback.

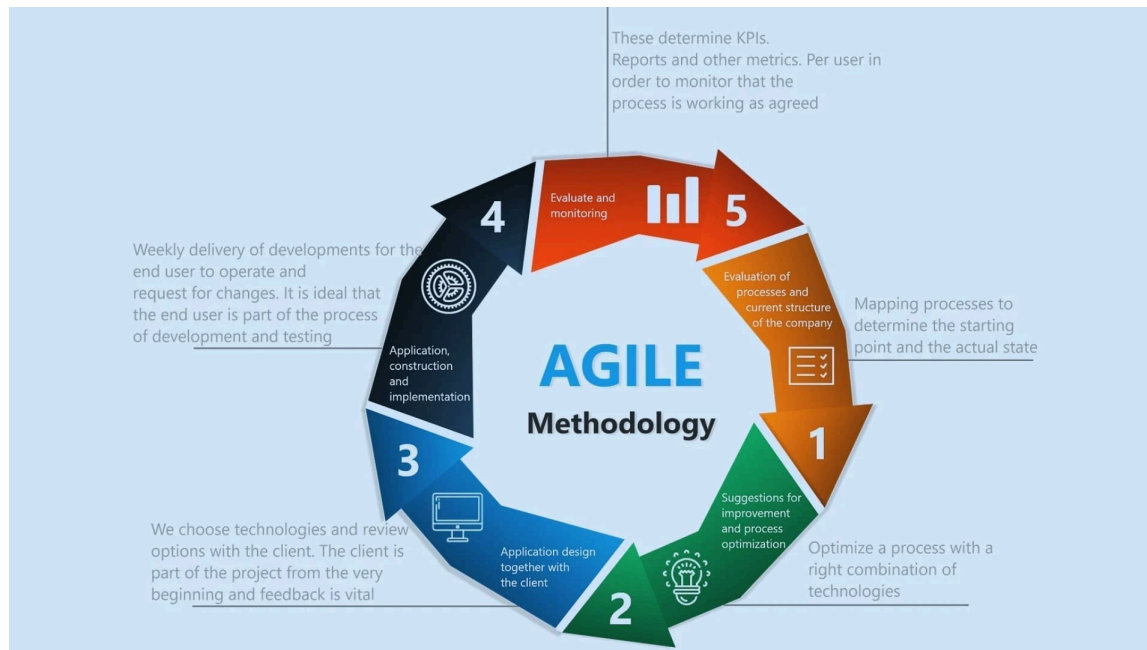
Why Agile?

Traditional models like Waterfall follow a strict sequence. Agile allows flexibility to change and adapt, leading to better end products and more satisfied users.

Agile Manifesto Principles

1. Individuals and interactions over processes and tools
2. Working software over comprehensive documentation
3. Customer collaboration over contract negotiation
4. Responding to change over following a plan

Agile Methodology



Key Components of Agile

1. Sprints - Time-boxed periods (1-4 weeks) for focused development
2. User Stories - Format: 'As a [user], I want [feature] so that [benefit]'
3. Product Backlog - List of prioritized tasks and features
4. Scrum Roles:
 - Product Owner – defines features and priorities
 - Scrum Master – facilitates team processes
 - Development Team – builds and tests the product
5. Core Meetings:
 - Daily Stand-up
 - Sprint Planning
 - Sprint Review

Agile Methodology

- Retrospective

Agile Frameworks

- Scrum: Sprint-based development with defined roles and ceremonies
- Kanban: Visual workflow using boards
- Extreme Programming (XP): Focuses on technical excellence
- Lean: Eliminates waste to improve efficiency

Agile Tools

- Jira, Trello, Asana for task management
- Slack for team communication

Agile vs Waterfall

Feature	Agile	Waterfall
Flexibility	High	Low
Testing	Continuous	After development
Delivery	Incremental	Final phase

Agile Methodology

User Involvement	Ongoing	Initial only
------------------	---------	--------------

Benefits

- Faster time to market
- High product quality
- Better stakeholder engagement
- Greater team collaboration

Challenges

- Hard to predict timelines/costs
- Needs skilled and disciplined teams
- May not suit all types of projects

Real-World Example

A startup developed a travel booking app using Agile. By showing progress every 2 weeks, the client gave real-time feedback. This helped the team fix issues quickly and deliver the final product faster with high quality.